Smart Campus Lost-and-Found Item Matching System

DBMS Experiential Learning – Level 3 Mini Project Report

1. Title Page

Field	Detail
Course Title	Database Management Systems (DBMS Experiential Learning – Level 3)
Course Code	UE22CS251A / CS251
Project Title	Smart Campus Lost-and-Found Item Matching System with Semantic Vector Retrieval
Course Instructors / Evaluators	Dr. Nagasundari S, Prof. Vinay Joshi
Academic Department	Department of Computer Science & Engineering, PES University
Repository URL	https://github.com/Uttam1916/DBMSMiniProject

Student Information & Responsibilities

Student Name	SRN	Role & Individual Contribution
Uttam K R	*PES1UG24CS697*	**Lead Developer & Database Architect (Sole Contributor)**: End-to-end design and implementation of both database and application layers. Conceptualized ER model, converted to 3NF relational schema, implemented PostgreSQL DDL/DML, PL/pgSQL triggers (`sync_item_status_after_claim` , `validate_claim_item` , `validate_item_match_types`), database views, recursive CTEs, and pgvector HNSW indexing. Built the complete Next.js 14 web application, FastEmbed ONNX 384-d vector embedding pipeline, ACID transaction handlers with row-level locking (`FOR UPDATE`), role-based access control (RBAC), and interactive evaluation tools.

2. Abstract

Campus lost-and-found operations typically suffer from vocabulary mismatch, manual record-keeping, and weak ownership verification. A student reporting a lost *"black stainless-steel bottle"* will not match a finder's record for a *"dark metal water bottle"* under standard keyword or SQL `ILIKE` pattern queries. Furthermore, traditional applications lack transactional concurrency control, allowing multiple conflicting claims on the same item.

This project implements a practical **DBMS Level 3 Experiential Learning System** combining relational integrity with vector similarity search in a single engine. Built with **PostgreSQL 16**, the **pgvector** extension, and dense text embeddings (`sentence-transformers/all-MiniLM-L6-v2` , 384 dimensions), the application provides:

1. **Semantic Vector Matching**: Maps natural language descriptions into 384-dimensional Euclidean space. Nearest-neighbor searches use cosine distance (`<=>`) backed by a Hierarchical Navigable Small World (**HNSW**) vector index. Potential matches are surfaced directly inside the lost item's detail view.
2. **ACID Transactions & Trigger Synchronization**: Manages claim adjudication using explicit database transactions with row-level locks (`FOR UPDATE`). When staff approves a claim, database trigger `sync_item_status_after_claim` automatically updates the item's status from `OPEN` to `CLAIMED` within the same transaction.
3. **Advanced SQL Concepts**: Features 3 PL/pgSQL triggers, 2 database views, a PL/pgSQL stored function (`find_matching_items`), 4-way joins, nested subqueries, correlated subqueries with `EXISTS` , Common Table Expressions (CTEs), and a recursive CTE (`WITH RECURSIVE`) traversing a campus zone hierarchy.
4. **Campus Web Application**: Next.js interface providing item browsing, multi-criteria filtering (category, location, date, type), item reporting with on-the-fly vectorization, ownership claim filing with proof, staff claim adjudication, and an evaluation console with millisecond query profiling.

3. Tech Stack & Technology Justification

3.1 Technology Stack Summary



- **Frontend**: Next.js 14 (App Router, TypeScript, Tailwind CSS, Lucide Icons).
- **Backend / API**: Next.js Route Handlers with direct connection pooling via `pg` (`node-postgres`).
- **Database Engine**: PostgreSQL 16 with native `pgvector` extension.
- **Embedding Model**: FastEmbed CPU runtime running `sentence-transformers/all-MiniLM-L6-v2` (384-dimensional dense vectors).
- **Containerization**: Docker (`pgvector/pgvector:pg16`).
- **Version Control**: Git & GitHub.

3.2 Technology Justification Matrix

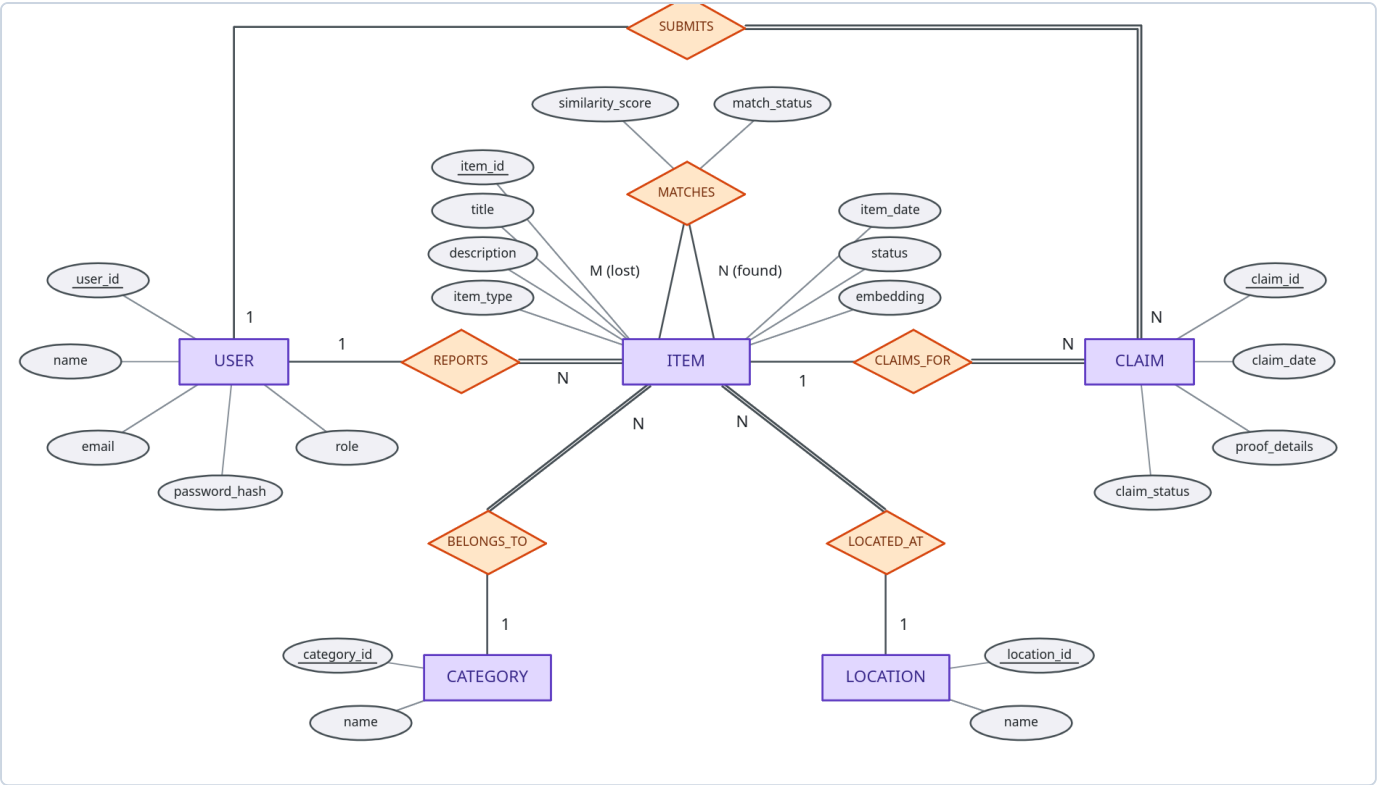
Requirement	Relational Only (MySQL)	Document NoSQL (MongoDB)	Dedicated Vector DB (Pinecone / Chroma)	Chosen: PostgreSQL 16 + pgvector
Referential Integrity	High (Foreign Keys, Cascade/Restrict)	Low (Application-level enforcement)	None (Key-value only)	**Highest (Foreign Keys, Cascade/Restrict, Engine-Level Checks)**
ACID Concurrency on Claims	Supported	Document-level only; multi-doc slow	Not supported	**Native Row-Level Locking (`FOR UPDATE`) & Atomic Blocks**
Vector Similarity Search	Unsupported / Primitive	Requires external Atlas Search	Supported	**Native `vector(384)` with HNSW index inside query engine**
Data Sync Latency	N/A	High (dual sync between DBs)	High (dual sync drift risks)	**Zero (Vectors reside directly within the `item` row)**
Complex Joins, Views & CTEs	Supported	Pipeline aggregations complex	Unsupported	**Native 4-way Joins, Views, CTEs, and Recursive Queries**
Operational Overhead	Medium (1 engine)	Medium (1 engine)	High (2 disparate databases to manage)	**Lowest (Single containerized PostgreSQL engine)**

Justification Summary:
Campus lost-and-found management requires relational integrity (guaranteeing that claims reference valid items and users cannot have duplicate emails) and transactional locking during claim adjudication. Simultaneously, solving vocabulary mismatch requires dense vector representations. Selecting **PostgreSQL 16 with pgvector** unifies relational ACID safety and vector indexing in a single operational engine without synchronization drift between two separate databases.

4. SQL - Database Commands

4.1 Entity-Relationship (ER) Diagram

The ER diagram defines 6 core entities: `USERS`, `CATEGORY`, `LOCATION`, `ITEM`, `ITEM_MATCH`, and `CLAIM`.



4.2 Relational Schema (3NF Normalization)

The relational schema converted to Third Normal Form (3NF) eliminates partial and transitive functional dependencies:



```
USERS (
    user_id BIGINT PRIMARY KEY GENERATED BY DEFAULT AS IDENTITY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(255) NOT NULL UNIQUE,
    password_hash TEXT NOT NULL,
    role VARCHAR(10) NOT NULL CHECK (role IN ('STUDENT', 'STAFF', 'ADMIN'))
);

CATEGORY (
    category_id INTEGER PRIMARY KEY GENERATED BY DEFAULT AS IDENTITY,
    name VARCHAR(100) NOT NULL UNIQUE
);

LOCATION (
    location_id INTEGER PRIMARY KEY GENERATED BY DEFAULT AS IDENTITY,
    name VARCHAR(150) NOT NULL UNIQUE,
    parent_location_id INTEGER REFERENCES LOCATION(location_id) ON DELETE SET NULL
);

ITEM (
    item_id BIGINT PRIMARY KEY GENERATED BY DEFAULT AS IDENTITY,
    title VARCHAR(200) NOT NULL,
    description TEXT NOT NULL,
    item_type VARCHAR(5) NOT NULL CHECK (item_type IN ('LOST', 'FOUND')),
    item_date DATE NOT NULL DEFAULT CURRENT_DATE,
    status VARCHAR(10) NOT NULL DEFAULT 'OPEN' CHECK (status IN ('OPEN', 'CLAIMED', 'RESOLVED', 'CLOSED')),
    embedding VECTOR(384),
    reported_by BIGINT NOT NULL REFERENCES USERS(user_id) ON DELETE RESTRICT,
    category_id INTEGER NOT NULL REFERENCES CATEGORY(category_id) ON DELETE RESTRICT,
    location_id INTEGER NOT NULL REFERENCES LOCATION(location_id) ON DELETE RESTRICT
);

ITEM_MATCH (
    lost_item_id BIGINT NOT NULL REFERENCES ITEM(item_id) ON DELETE CASCADE,
    found_item_id BIGINT NOT NULL REFERENCES ITEM(item_id) ON DELETE CASCADE,
    similarity_score DOUBLE PRECISION NOT NULL CHECK (similarity_score BETWEEN -1 AND 1),
    match_status VARCHAR(10) NOT NULL DEFAULT 'SUGGESTED' CHECK (match_status IN ('SUGGESTED', 'ACCEPTED', 'REJECTED')),
    PRIMARY KEY (lost_item_id, found_item_id),
    CHECK (lost_item_id <> found_item_id)
);

CLAIM (
    claim_id BIGINT PRIMARY KEY GENERATED BY DEFAULT AS IDENTITY,
    claim_date DATE NOT NULL DEFAULT CURRENT_DATE,
    proof_details TEXT NOT NULL,
    claim_status VARCHAR(10) NOT NULL DEFAULT 'PENDING' CHECK (claim_status IN ('PENDING', 'APPROVED', 'REJECTED')),
    claimed_by BIGINT NOT NULL REFERENCES USERS(user_id) ON DELETE RESTRICT,
    item_id BIGINT NOT NULL REFERENCES ITEM(item_id) ON DELETE RESTRICT
);
```

4.3 DDL Implementation & Constraints

```
-- DDL ALTER Demonstration: Expand title capacity as taxonomy requirements evolved
ALTER TABLE item ALTER COLUMN title TYPE VARCHAR(200);

-- Indexes for relational filtering and foreign key joins
CREATE INDEX idx_item_type_status ON item(item_type, status);
CREATE INDEX idx_item_reported_by ON item(reported_by);
CREATE INDEX idx_item_category ON item(category_id);
CREATE INDEX idx_item_location ON item(location_id);
CREATE INDEX idx_claim_item ON claim(item_id);
CREATE INDEX idx_claim_claimed_by ON claim(claimed_by);
CREATE INDEX idx_item_match_found ON item_match(found_item_id);

-- Partial Unique Index: Guarantees at most one approved claim per item
CREATE UNIQUE INDEX idx_one_approved_claim_per_item
    ON claim(item_id) WHERE claim_status = 'APPROVED';
```

4.4 Advanced SQL: PL/pgSQL Triggers

Trigger 1: Automatic Status Synchronization on Claim Approval

```
CREATE OR REPLACE FUNCTION sync_item_status_after_claim()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
BEGIN
    IF NEW.claim_status = 'APPROVED' THEN
        IF TG_OP = 'INSERT' OR OLD.claim_status IS DISTINCT FROM 'APPROVED' THEN
            UPDATE item SET status = 'CLAIMED'
            WHERE item_id = NEW.item_id AND status = 'OPEN';
            IF NOT FOUND THEN
                RAISE EXCEPTION 'Only an OPEN item can have an approved claim';
            END IF;
        END IF;
    ELSIF TG_OP = 'UPDATE' AND OLD.claim_status = 'APPROVED' AND NEW.claim_status <> 'APPROVED' THEN
        UPDATE item SET status = 'OPEN'
        WHERE item_id = OLD.item_id AND status = 'CLAIMED';
    END IF;
    RETURN NEW;
END;
$$;

CREATE TRIGGER trg_claim_update_sync_item_status
AFTER UPDATE OF claim_status ON claim
FOR EACH ROW EXECUTE FUNCTION sync_item_status_after_claim();
```

Trigger 2: Target Item Validation for Claims

```
CREATE OR REPLACE FUNCTION validate_claim_item()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
DECLARE
    current_type   VARCHAR(5);
    current_status VARCHAR(10);
BEGIN
    IF TG_OP = 'UPDATE' THEN
        IF NEW.item_id IS DISTINCT FROM OLD.item_id THEN
            RAISE EXCEPTION 'An existing claim cannot be moved to another item';
        END IF;
        RETURN NEW;
    END IF;

    SELECT item_type, status INTO current_type, current_status
    FROM item WHERE item_id = NEW.item_id;

    IF NOT FOUND THEN
        RAISE EXCEPTION 'The claimed item does not exist';
    END IF;
    IF current_type IS DISTINCT FROM 'FOUND' THEN
        RAISE EXCEPTION 'Claims can only be submitted for FOUND items';
    END IF;
    IF current_status IS DISTINCT FROM 'OPEN' THEN
        RAISE EXCEPTION 'Claims can only be submitted for OPEN items';
    END IF;
    RETURN NEW;
END;
$$;

CREATE TRIGGER trg_validate_claim_item
BEFORE INSERT OR UPDATE ON claim
FOR EACH ROW EXECUTE FUNCTION validate_claim_item();
```

Trigger 3: Match Pairing Integrity

```
CREATE OR REPLACE FUNCTION validate_item_match_types()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
DECLARE
    lost_type  VARCHAR(5);
    found_type VARCHAR(5);
BEGIN
    SELECT item_type INTO lost_type FROM item WHERE item_id = NEW.lost_item_id;
    SELECT item_type INTO found_type FROM item WHERE item_id = NEW.found_item_id;

    IF lost_type IS DISTINCT FROM 'LOST' THEN
        RAISE EXCEPTION 'lost_item_id must refer to an item whose item_type is LOST';
    END IF;
    IF found_type IS DISTINCT FROM 'FOUND' THEN
        RAISE EXCEPTION 'found_item_id must refer to an item whose item_type is FOUND';
    END IF;
    RETURN NEW;
END;
$$;

CREATE TRIGGER trg_validate_item_match_types
BEFORE INSERT OR UPDATE ON item_match
FOR EACH ROW EXECUTE FUNCTION validate_item_match_types();
```

4.5 Advanced SQL: Database Views

```
-- View 1: Unified item catalog with reporter, category, and location projections
CREATE VIEW item_details_view AS
SELECT i.item_id, i.title, i.description, i.item_type, i.item_date, i.status,
       u.name AS reporter_name, c.name AS category_name, l.name AS location_name
FROM item i
JOIN users u ON u.user_id = i.reported_by
JOIN category c ON c.category_id = i.category_id
JOIN location l ON l.location_id = i.location_id;

-- View 2: Pre-joined administrative review queue for pending claims
CREATE VIEW pending_claims_view AS
SELECT cl.claim_id, cl.claim_date, cl.proof_details,
       i.item_id, i.title AS item_title, u.user_id AS claimant_id,
       u.name AS claimant_name, cl.claim_status
FROM claim cl
JOIN item i ON i.item_id = cl.item_id
JOIN users u ON u.user_id = cl.claimed_by
WHERE cl.claim_status = 'PENDING';
```

4.6 Advanced SQL: PL/pgSQL Stored Function

```
CREATE OR REPLACE FUNCTION find_matching_items(
    p_lost_item_id BIGINT,
    p_limit INTEGER DEFAULT 5
)
RETURNS TABLE (
    found_item_id    BIGINT,
    title            TEXT,
    description      TEXT,
    similarity_score DOUBLE PRECISION
)
LANGUAGE plpgsql STABLE AS $$
BEGIN
    IF p_limit < 1 OR p_limit > 50 THEN
        RAISE EXCEPTION 'p_limit must be between 1 and 50';
    END IF;

    RETURN QUERY
    SELECT f.item_id,
           f.title::TEXT,
           f.description,
           1 - (lost.embedding <=> f.embedding) AS similarity_score
    FROM item lost
    JOIN item f ON f.item_type = 'FOUND'
    WHERE lost.item_id = p_lost_item_id
      AND lost.item_type = 'LOST'
      AND lost.embedding IS NOT NULL
      AND f.embedding IS NOT NULL
      AND f.status = 'OPEN'
    ORDER BY lost.embedding <=> f.embedding
    LIMIT p_limit;
END;
$$;
```

4.7 DML Operations: INSERT, UPDATE, DELETE, and Multi-Table SELECT

```
-- DML INSERT with dynamic foreign key lookups
INSERT INTO item (title, description, item_type, item_date, status, reported_by, category_id, location_id)
SELECT 'Black stainless-steel bottle', 'Black bottle with a silver cap and a small sticker',
       'LOST', CURRENT_DATE - 2, 'OPEN', u.user_id, c.category_id, l.location_id
FROM users u CROSS JOIN category c CROSS JOIN location l
WHERE u.email = 'student@campus.edu' AND c.name = 'Personal Items' AND l.name = 'Library';

-- DML UPDATE with status mutation
UPDATE item SET status = 'RESOLVED' WHERE item_id = 1;

-- DML DELETE with RETURNING clause (cascades to item_match)
DELETE FROM item WHERE title = 'Temporary test umbrella' RETURNING item_id, title;

-- Multi-Table 4-Way Relational JOIN
SELECT i.item_id, i.title, i.item_type, i.status,
       c.name AS category, l.name AS location, u.name AS reported_by, i.item_date
FROM item i
JOIN category c ON i.category_id = c.category_id
JOIN location l ON i.location_id = l.location_id
JOIN users u ON i.reported_by = u.user_id
ORDER BY i.item_date DESC, i.item_id DESC LIMIT 10;
```

4.8 Nested, Correlated, and Recursive Queries

Nested Subquery with `IN` and Grouped `HAVING`

```
SELECT user_id, name, email, role
FROM users
WHERE user_id IN (
    SELECT claimed_by FROM claim GROUP BY claimed_by HAVING COUNT(*) >= 1
);
```

Correlated Subquery with `EXISTS`

```
SELECT
    i.item_id,
    i.title,
    i.item_type,
    i.status,
    (SELECT COUNT(*) FROM claim c WHERE c.item_id = i.item_id) AS total_claims_filed
FROM item i
WHERE i.item_type = 'FOUND'
  AND EXISTS (SELECT 1 FROM claim c WHERE c.item_id = i.item_id)
ORDER BY total_claims_filed DESC;
```

Common Table Expression (CTE): Match Confidence Tiering

```
WITH item_stats AS (
    SELECT lost_item_id, MAX(similarity_score) AS best_score, COUNT(*) AS candidate_count
    FROM item_match GROUP BY lost_item_id
),
match_categorized AS (
    SELECT s.lost_item_id, i.title AS lost_title, ROUND(s.best_score::numeric, 4) AS best_similarity,
           s.candidate_count,
           CASE
               WHEN s.best_score >= 0.70 THEN 'HIGH_CONFIDENCE'
               WHEN s.best_score >= 0.50 THEN 'MEDIUM_CONFIDENCE'
               ELSE 'LOW_CONFIDENCE'
           END AS match_tier
    FROM item_stats s JOIN item i ON i.item_id = s.lost_item_id
)
SELECT * FROM match_categorized ORDER BY best_similarity DESC;
```

Recursive Query (`WITH RECURSIVE`): Campus Location Hierarchy Graph Traversal

```
WITH RECURSIVE location_hierarchy AS (
    -- Anchor Member: Root campus zones (parent_location_id IS NULL)
    SELECT location_id, name, parent_location_id, 1 AS depth, name::TEXT AS path
    FROM location WHERE parent_location_id IS NULL

    UNION ALL

    -- Recursive Member: Child zones linked to their parent
    SELECT c.location_id, c.name, c.parent_location_id, lh.depth + 1, (lh.path || ' -> ' || c.name)::TEXT
    FROM location c JOIN location_hierarchy lh ON c.parent_location_id = lh.location_id
)
SELECT lh.location_id, lh.path AS zone_hierarchy, lh.depth, COUNT(i.item_id) AS items_reported
FROM location_hierarchy lh
LEFT JOIN item i ON i.location_id = lh.location_id
GROUP BY lh.location_id, lh.path, lh.depth
ORDER BY lh.depth, lh.location_id;
```

4.9 ACID Transaction Adjudication

```
BEGIN;

-- Lock item and claim records concurrently
SELECT cl.claim_id, cl.claim_status, i.item_id, i.title, i.status AS item_status_before
FROM claim cl
JOIN item i ON i.item_id = cl.item_id
WHERE cl.claim_id = 1
FOR UPDATE;

-- Update claim status (Trigger sync_item_status_after_claim fires atomically)
UPDATE claim SET claim_status = 'APPROVED' WHERE claim_id = 1;

-- Verify item status transitioned to CLAIMED within the transaction
SELECT cl.claim_id, cl.claim_status, i.item_id, i.title, i.status AS item_status_after
FROM claim cl
JOIN item i ON i.item_id = cl.item_id
WHERE cl.claim_id = 1;

COMMIT;
```

5. NoSQL / Vector Database Commands

5.1 Vector Database Creation & Extensions

```
-- Enable vector extension
CREATE EXTENSION IF NOT EXISTS vector;

-- Embedding attribute definition inside ITEM table
-- Data type: VECTOR(384) matching sentence-transformers/all-MiniLM-L6-v2
ALTER TABLE item ADD COLUMN IF NOT EXISTS embedding VECTOR(384);

-- Hierarchical Navigable Small World (HNSW) Index
-- Uses vector_cosine_ops for cosine similarity search
CREATE INDEX IF NOT EXISTS idx_item_embedding_hnsw
    ON item USING hnsw (embedding vector_cosine_ops)
    WHERE embedding IS NOT NULL;
```

5.2 Vector CRUD Operations

1. Create (Insert Item with Pre-Computed Vector)

```
INSERT INTO item (item_id, title, description, item_type, item_date, status, reported_by, category_id, location_id, embedding)
VALUES (
    1,
    'Black stainless-steel bottle',
    'Black bottle with a silver cap and a small sticker',
    'LOST',
    CURRENT_DATE,
    'OPEN',
    1, 4, 1,
    '[-0.110855, 0.068705, -0.041783, 0.054057, 0.074294, ...]'::vector
);
```

2. Read (Cosine Distance Similarity Retrieval)

```
-- Find top 5 FOUND items nearest to an ad-hoc query vector
SELECT
    item_id,
    title,
    description,
    ROUND((1 - (embedding <=> '[-0.1346, 0.0534, ...]'::vector))::numeric, 4) AS similarity_score
FROM item
WHERE item_type = 'FOUND' AND status = 'OPEN' AND embedding IS NOT NULL
ORDER BY embedding <=> '[-0.1346, 0.0534, ...]'::vector
LIMIT 5;
```

3. Update (Automated Pairings Refresh via Lateral Join)

```
INSERT INTO item_match (lost_item_id, found_item_id, similarity_score, match_status)
SELECT lost.item_id, candidate.found_item_id, candidate.similarity_score, 'SUGGESTED'
FROM item lost
CROSS JOIN LATERAL find_matching_items(lost.item_id, 5) candidate
WHERE lost.item_type = 'LOST'
ON CONFLICT (lost_item_id, found_item_id)
DO UPDATE SET similarity_score = EXCLUDED.similarity_score, match_status = 'SUGGESTED';
```

4. Delete (Cascade Cleanup of Vector Pairings)

```
-- Deleting an item automatically deletes dependent vector pairings in item_match
DELETE FROM item WHERE item_id = 1 RETURNING item_id, title;
```

5.3 Embedding Model Architecture & Strategy

- **Model**: `sentence-transformers/all-MiniLM-L6-v2` executed via ONNX Runtime CPU (`FastEmbed`).
- **Dimensionality**: 384 dimensions normalized float vectors.
- **Embedded Text**: Concatenation of item title and detailed description (`f"{title}. {description}"`).
- **Metric**: Cosine distance ($d = 1 - \cos(\theta)$), accessed via PostgreSQL operator `<=>`.
- **Similarity Formula**: $\text{similarity_score} = 1 - (\mathbf{u} \Leftrightarrow \mathbf{v})$.

6. Next.js - Project Tree Structure

```
DBMSmini/
├── .env.example                        # Environment configuration template
├── .env.local                          # Local PostgreSQL connection parameters
├── .gitignore                          # Git ignore rules
├── campus_lost_found_setup.sql         # Self-contained master SQL setup (DDL, triggers, views, data)
├── demo_queries.sql                    # All 15 demonstration & evaluation queries
├── next.config.mjs                     # Next.js configuration
├── package.json                        # Node dependencies & execution scripts
├── package-lock.json                   # Dependency lockfile
├── postcss.config.mjs                  # Tailwind CSS PostCSS plugin
├── PROJECT_REPORT.md                   # Formal Academic Level 3 Report (this document)
├── public/
│   ├── lost_found_ER.png               # Public asset ER diagram
│   └── lost_found_relational_schema.png # Public asset Relational diagram
├── README.md                           # Setup guide, execution manual, and demo walkthrough
├── scripts/
│   └── embed_query.py                  # FastEmbed ONNX 384-d vector generation helper
├── src/
│   ├── app/
│   │   ├── api/
│   │   │   ├── analytics/route.ts      # Aggregated category metrics, trigger/view status
│   │   │   ├── claims/
│   │   │   │   ├── [id]/route.ts       # ACID claim approval/rejection with RBAC check
│   │   │   │   └── route.ts            # Claims catalog and submission with trigger check
│   │   │   ├── demo/reset/route.ts     # One-click demo database restoration
│   │   │   ├── items/
│   │   │   │   ├── [id]/route.ts       # Single item details, cascade delete, status update
│   │   │   │   └── route.ts            # Filtered items list & auto-vectorized item insert
│   │   │   ├── matches/route.ts        # Suggested pairings from item_match & lateral refresh
│   │   │   ├── meta/route.ts           # Categories, locations hierarchy, and demo users
│   │   │   ├── queries/run/route.ts    # Pre-packaged evaluation queries + custom runner
│   │   │   └── search/semantic/route.ts # Live side-by-side vector vs keyword search
│   │   ├── globals.css                 # Custom styling and typography
│   │   ├── layout.tsx                  # Root HTML structure and metadata
│   │   └── page.tsx                    # Main client page, session management, and routing
│   ├── components/
│   │   ├── BrowseView.tsx              # Central catalog with multi-column filters & search
│   │   ├── MyActivityView.tsx          # Student/Staff view of their reported items & claims
│   │   ├── ClaimsReviewView.tsx        # Staff/Admin claim review queue with proof inspection
│   │   ├── Header.tsx                  # Top navigation with profile switcher & action button
│   │   ├── ItemDetailModal.tsx         # Item deep inspector with unconfirmed potential matches
│   │   ├── ReportItemModal.tsx         # Item submission modal with FastEmbed vectorization
│   │   ├── FileClaimModal.tsx          # Claim filing modal with trigger validation
│   │   ├── SignInModal.tsx             # Campus authentication and account switcher
│   │   └── SystemToolsModal.tsx        # Evaluation console with SQL runner, schema, and reset
│   ├── lib/
│   │   ├── db.ts                       # Connection pool singleton (`pg`) with transaction helper
│   │   └── embed.ts                    # FastEmbed generation wrapper and match refresher
│   └── types/
│       └── index.ts                    # TypeScript data interfaces
├── tailwind.config.ts                  # Tailwind configuration
└── tsconfig.json                       # TypeScript compiler options
```

6.1 Backend API Route Responsibilities

1. **`GET / POST /api/items`**:
   - `GET`: Supports multi-predicate relational filtering (`type`, `category_id`, `location_id`, `status`, `search`, `reported_by`, `date_filter`).
   - `POST`: Validates inputs, calls `generateTextEmbedding()` to compute a 384-d vector via ONNX, saves item into PostgreSQL, and executes `refreshItemMatches()` to update candidate pairings.
2. **`PATCH /api/claims/[id]`**:
   - Adjudicates claims inside an explicit `BEGIN ... COMMIT` block.
   - Enforces RBAC: verifies `user_role` is `STAFF` or `ADMIN` (rejects students with 403 Forbidden).
   - Fires trigger `sync_item_status_after_claim`, returning execution proof to the UI.
3. **`POST /api/search/semantic`**:
   - Computes query vector and executes cosine similarity query (`1 - (embedding <=> $1)`).
   - Simultaneously runs relational `ILIKE` query across titles and descriptions.
   - Returns both result sets side-by-side for TA comparison.
4. **`POST /api/queries/run`**:
   - Pre-loaded with all Level 3 evaluation queries.

- Profiles execution time in milliseconds (`performance.now()`) and returns formatted column headers and data rows.

5. `POST /api/demo/reset`:
   - Restores the database back to clean baseline demo records in 1 second.

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7. Few Sample Screenshots of Application Code & Demo

7.1 Application Code Implementation Screenshots

Figure 7.1: Core PostgreSQL DDL with `VECTOR(384)` dense embedding attribute, partial unique index `idx_one_approved_claim_per_item`, HNSW cosine index, and PL/pgSQL trigger `sync_item_status_after_claim`.

```sql
-- Table Definition with 384-dimensional Vector Attribute
CREATE TABLE item (
    item_id     BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    title       VARCHAR(200) NOT NULL,
    description TEXT NOT NULL,
    item_type   VARCHAR(5) NOT NULL CHECK (item_type IN ('LOST', 'FOUND')),
    item_date   DATE NOT NULL DEFAULT CURRENT_DATE,
    status      VARCHAR(10) NOT NULL DEFAULT 'OPEN'
                CHECK (status IN ('OPEN', 'CLAIMED', 'RESOLVED', 'CLOSED')),
    embedding   VECTOR(384),
    reported_by BIGINT NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    category_id INTEGER NOT NULL REFERENCES category(category_id) ON DELETE RESTRICT,
    location_id INTEGER NOT NULL REFERENCES location(location_id) ON DELETE RESTRICT
);

-- Partial Unique Index: Exactly one approved claim per item
CREATE UNIQUE INDEX idx_one_approved_claim_per_item
    ON claim(item_id) WHERE claim_status = 'APPROVED';

-- HNSW Cosine Similarity Index on Dense Vector Embeddings
CREATE INDEX idx_item_embedding_hnsw
    ON item USING hnsw (embedding vector_cosine_ops)
    WITH (m = 16, ef_construction = 64);

-- Trigger: Atomic Status Mutation inside BEGIN...COMMIT
CREATE OR REPLACE FUNCTION sync_item_status_after_claim()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
BEGIN
    IF NEW.claim_status = 'APPROVED' THEN
        IF TG_OP = 'INSERT' OR OLD.claim_status IS DISTINCT FROM 'APPROVED' THEN
            UPDATE item SET status = 'CLAIMED'
            WHERE item_id = NEW.item_id AND status = 'OPEN';
            IF NOT FOUND THEN
                RAISE EXCEPTION 'Only an OPEN item can have an approved claim';
            END IF;
        END IF;
    ELSIF TG_OP = 'UPDATE' AND OLD.claim_status = 'APPROVED' AND NEW.claim_status <> 'APPROVED' THEN
        UPDATE item SET status = 'OPEN'
        WHERE item_id = OLD.item_id AND status = 'CLAIMED';
```

Code Explanation: The table definition establishes strong domain integrity using `CHECK` constraints on `item_type` and `status`, foreign keys with `ON DELETE RESTRICT` to prevent orphan cascades, and a native `VECTOR(384)` column for dense semantic embeddings. A partial unique index (`idx_one_approved_claim_per_item`) guarantees at the database engine level that at most one claim per item can reach `APPROVED` status. An HNSW cosine index (`m=16, ef_construction=64`) accelerates high-dimensional nearest-neighbor retrieval, while trigger function `sync_item_status_after_claim` automatically transitions item state between `OPEN` and `CLAIMED` when claims are adjudicated.

Figure 7.2: Vector embedding generation pipeline using FastEmbed ONNX runtime (`sentence-transformers/all-MiniLM-L6-v2`) and automated lateral match refresh (`refreshItemMatches`).

```typescript
// Dense 384-d vector generation via FastEmbed ONNX runtime
export async function generateTextEmbedding(text: string): Promise<number[]> {
  const scriptPath = path.join(process.cwd(), 'scripts', 'embed_query.py');
  return new Promise((resolve, reject) => {
    const python = spawn('python3', [scriptPath, text]);
    let stdout = '';
    let stderr = '';

    python.stdout.on('data', (d) => { stdout += d.toString(); });
    python.stderr.on('data', (d) => { stderr += d.toString(); });

    python.on('close', (code) => {
      if (code !== 0) return reject(new Error(`FastEmbed failed: ${stderr}`));
      try {
        const vector = JSON.parse(stdout.trim());
        if (!Array.isArray(vector) || vector.length !== 384) {
          return reject(new Error('Invalid 384-d vector output'));
        }
        resolve(vector);
      } catch (e) {
        reject(new Error(`Failed to parse embedding: ${stdout}`));
      }
    });
  });
}

// Automated lateral refresh of potential item matches
export async function refreshItemMatches(itemId: number) {
  const pool = getPool();
  await pool.query(
    `INSERT INTO item_match (lost_item_id, found_item_id, similarity_score, match_status)
     SELECT f.lost_id, f.found_id, f.similarity, 'SUGGESTED'
     FROM item i
     CROSS JOIN LATERAL find_matching_items(i.item_id, 5) f
     WHERE i.item_id = $1 AND i.item_type = 'LOST'
     ON CONFLICT (lost_item_id, found_item_id)
     DO UPDATE SET similarity_score = EXCLUDED.similarity_score;`,
    [itemId]
  );
```

Code Explanation: The `generateTextEmbedding` function interfaces with a local FastEmbed ONNX runtime executing `sentence-transformers/all-MiniLM-L6-v2` on CPU, converting unstructured lost-and-found text descriptions into normalized 384-dimensional dense vectors without external cloud API latency. Upon item insertion, `refreshItemMatches` executes a `CROSS JOIN LATERAL` query against stored PL/pgSQL function `find_matching_items(itemId, 5)`, computing cosine distances and populating candidate pairings in `item_match` with `ON CONFLICT DO UPDATE` idempotency.

Figure 7.3: ACID claim adjudication route (`/api/claims/[id]`) showing row-level locking (`FOR UPDATE`), role-based access control (RBAC), and trigger execution verification.

```
// Role-Based Access Control: Block students from claim adjudication
if (user_role === 'STUDENT') {
  return NextResponse.json(
    { error: 'Forbidden: Students cannot adjudicate claims. Staff or Admin role required.' },
    { status: 403 }
  );
}

// Dedicated client for explicit ACID transaction with row-level locking
const client = await getClient();
try {
  await client.query('BEGIN;');

  // Inspect item state before update with row-level lock (FOR UPDATE)
  const beforeRes = await client.query(
    `SELECT cl.claim_id, cl.item_id, i.title, i.status AS item_status
     FROM claim cl
     JOIN item i ON i.item_id = cl.item_id
     WHERE cl.claim_id = $1 FOR UPDATE;`,
    [claimId]
  );

  // Update claim: Trigger sync_item_status_after_claim fires automatically
  await client.query(
    `UPDATE claim SET claim_status = $1 WHERE claim_id = $2;`,
    [claim_status, claimId]
  );

  // Read post-trigger item state within the same transaction
  const afterRes = await client.query(
    `SELECT item_id, title, status AS item_status FROM item WHERE item_id = $1;`,
    [beforeRes.rows[0].item_id]
  );

  await client.query('COMMIT;');
  return NextResponse.json({
    message: `Claim status updated to ${claim_status}`,
    transaction_details: {
      item_status_before: beforeRes.rows[0].item_status,
```

Code Explanation: This endpoint enforces role-based access control at the server boundary, returning `403 Forbidden` if a student attempts claim adjudication. A dedicated database client executes within an explicit `BEGIN ... COMMIT` transaction, acquiring an exclusive row-level lock (`FOR UPDATE`) on the claim and parent item to prevent concurrent approval race conditions. When the claim update executes, trigger `sync_item_status_after_claim` fires atomically, mutating the item's status to `CLAIMED` before committing.

Figure 7.4: Semantic vector retrieval API handler comparing pgvector cosine distance (`1 - (i.embedding <=> $1::vector)`) against traditional relational `ILIKE` pattern matching.

```ts
// POST /api/search/semantic
export async function POST(req: NextRequest) {
  const { query, limit = 5 } = await req.json();
  const pool = getPool();

  // 1. Generate 384-d dense vector for user query
  const embedding = await generateTextEmbedding(query);
  const vectorStr = `[${embedding.join(',')}]`;

  // 2. Vector Cosine Similarity Search using pgvector (<=> operator)
  const vectorRes = await pool.query(
    `SELECT i.item_id, i.title, i.description, i.item_type, i.status,
            ROUND((1 - (i.embedding <=> $1::vector))::numeric, 4) AS similarity_score
     FROM item i
     WHERE i.status = 'OPEN' AND i.embedding IS NOT NULL
     ORDER BY i.embedding <=> $1::vector ASC
     LIMIT $2;`,
    [vectorStr, limit]
  );

  // 3. Traditional Keyword Search (ILIKE) across title and description
  const keywordRes = await pool.query(
    `SELECT i.item_id, i.title, i.description, i.item_type, i.status
     FROM item i
     WHERE i.status = 'OPEN'
       AND (i.title ILIKE $1 OR i.description ILIKE $1)
     LIMIT $2;`,
    [`%${query}%`, limit]
  );

  return NextResponse.json({
    vector_results: vectorRes.rows,
    keyword_results: keywordRes.rows,
    comparison: {
      query,
      vector_count: vectorRes.rows.length,
      keyword_count: keywordRes.rows.length
    }
  });
```

Code Explanation: The route accepts a natural language search query, generates its 384-dimensional dense vector representation, and executes a pgvector cosine similarity query (`1 - (embedding <=> $1::vector)`) against open items. Concurrently, it runs a relational `ILIKE` pattern match across `title` and `description`. Returning both result sets side-by-side demonstrates how semantic vector retrieval overcomes vocabulary mismatch where traditional keyword matching returns zero hits.

7.2 Live Application Demonstration Screenshots

Figure 7.5: Campus lost-and-found catalog with real-time multi-predicate filters (item type, category, location, date range, and status).

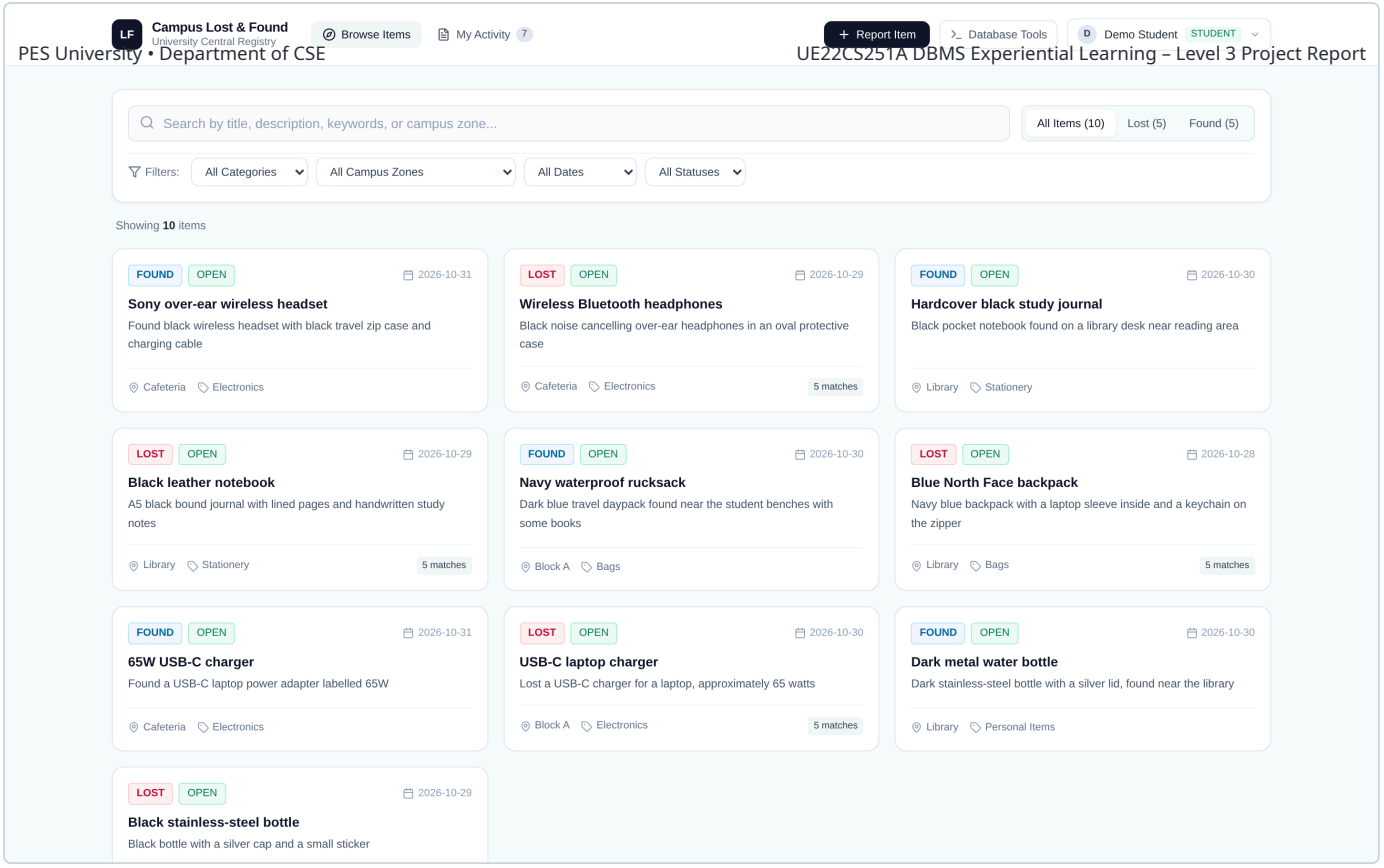


Figure 7.6: Item details view with semantic matching drawer displaying unconfirmed potential matches with cosine similarity scores.

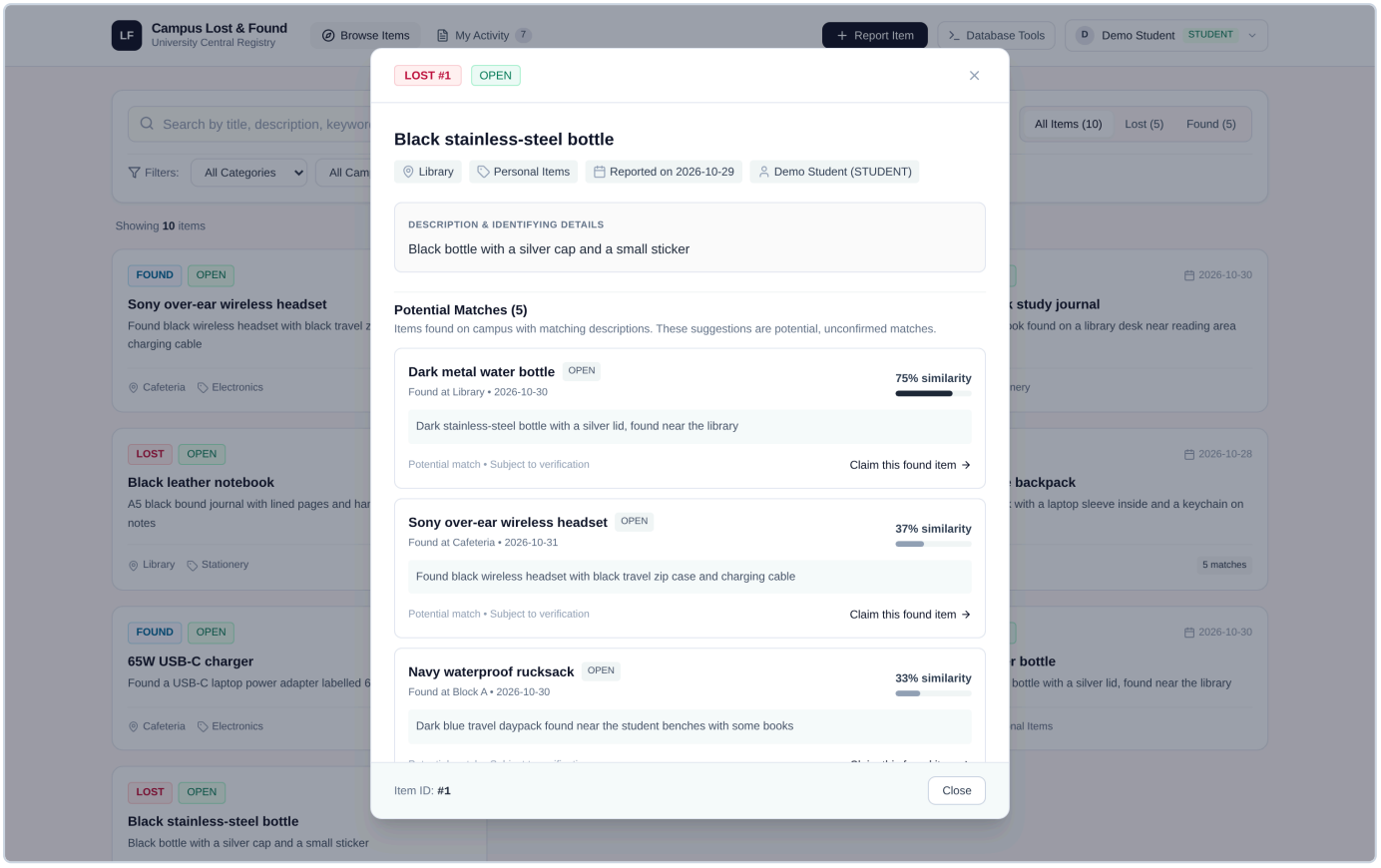


Figure 7.7: Item reporting interface featuring on-the-fly dense vector embedding computation upon submission.

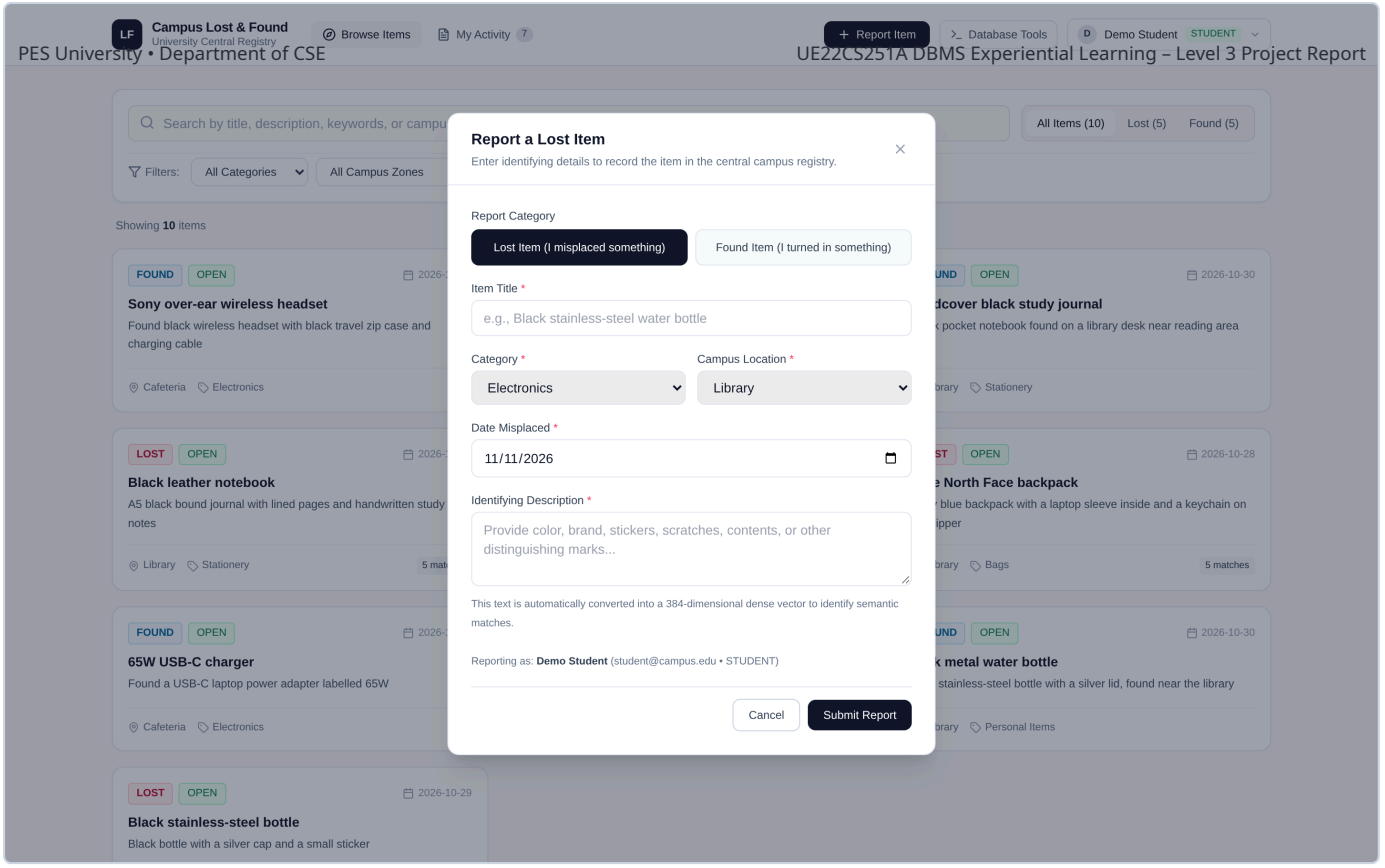


Figure 7.8: Student personal workspace showing reported items with status indicators and submitted ownership claims.

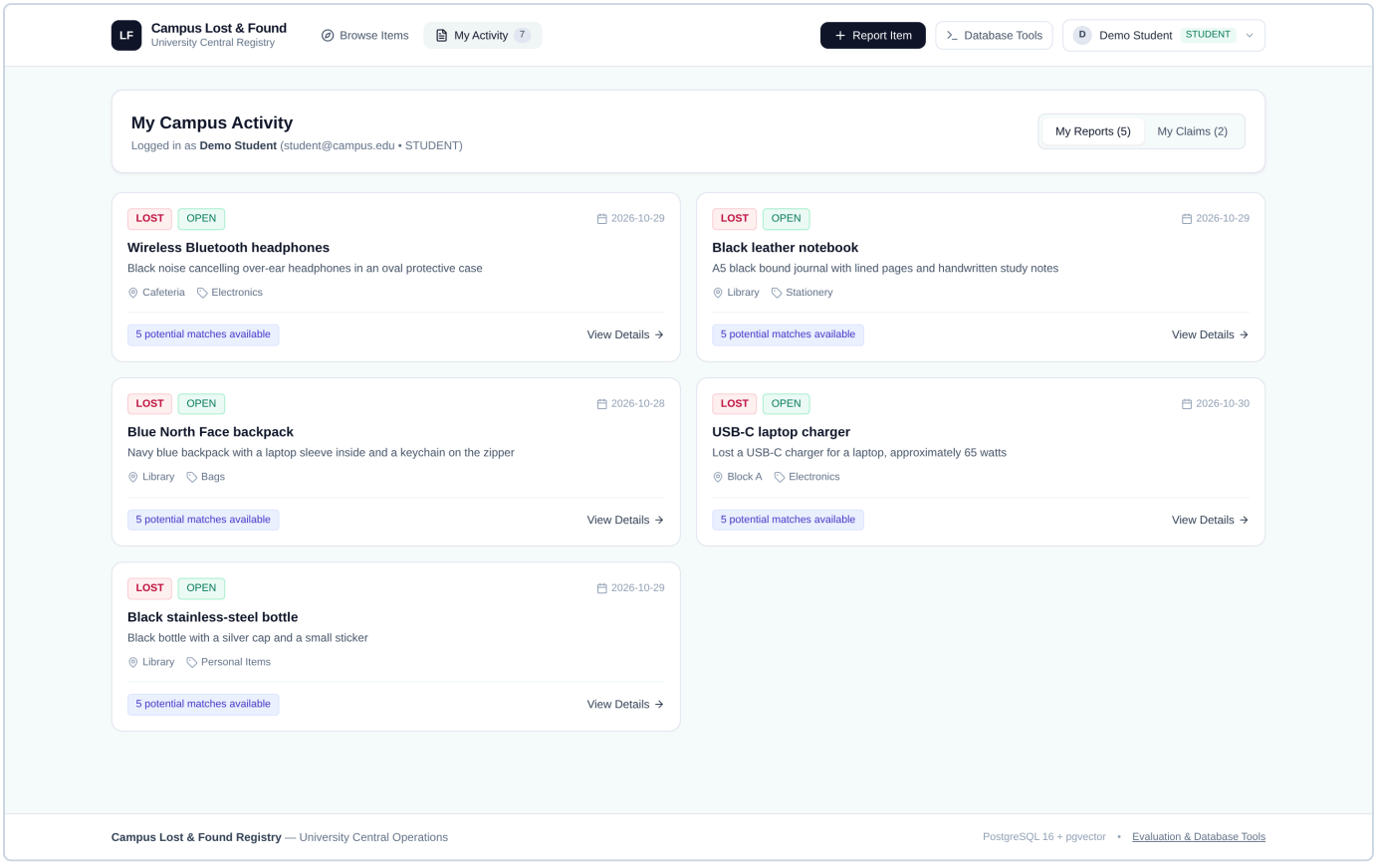


Figure 7.9: Staff claims adjudication queue displaying claimant proof details and transactional confirmation of trigger execution.

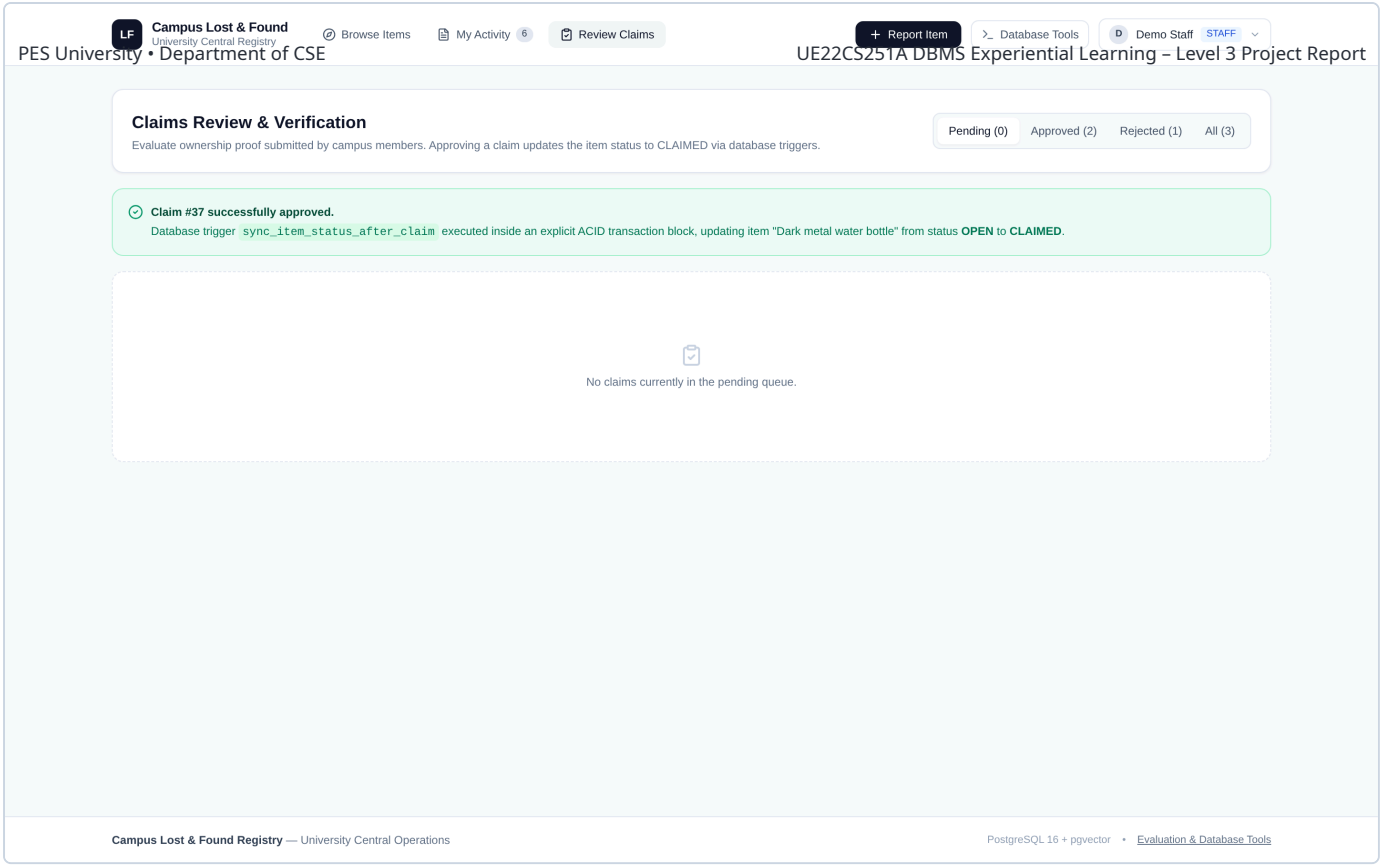


Figure 7.10: Interactive SQL evaluation console executing recursive campus location hierarchy query with millisecond profiling.

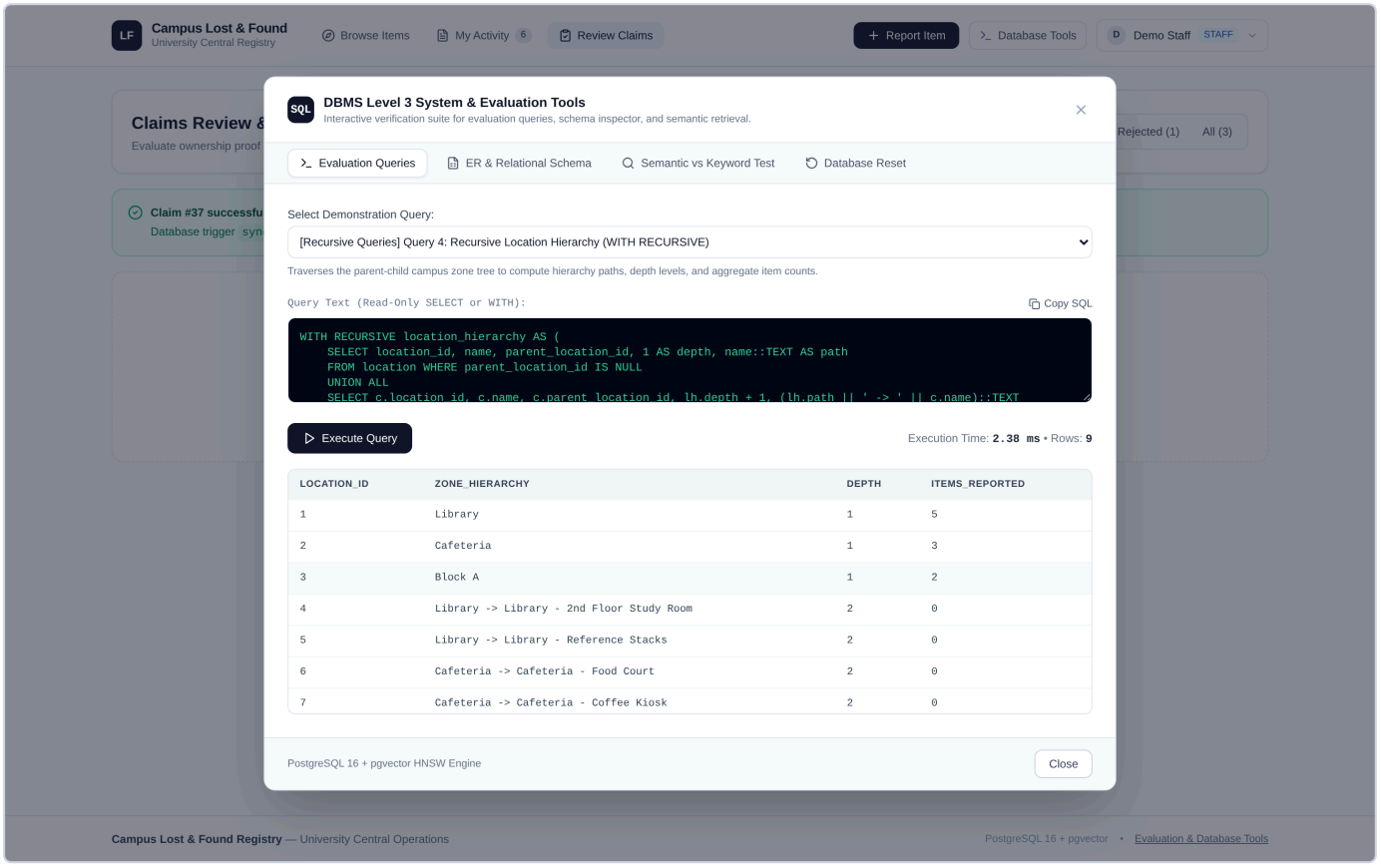
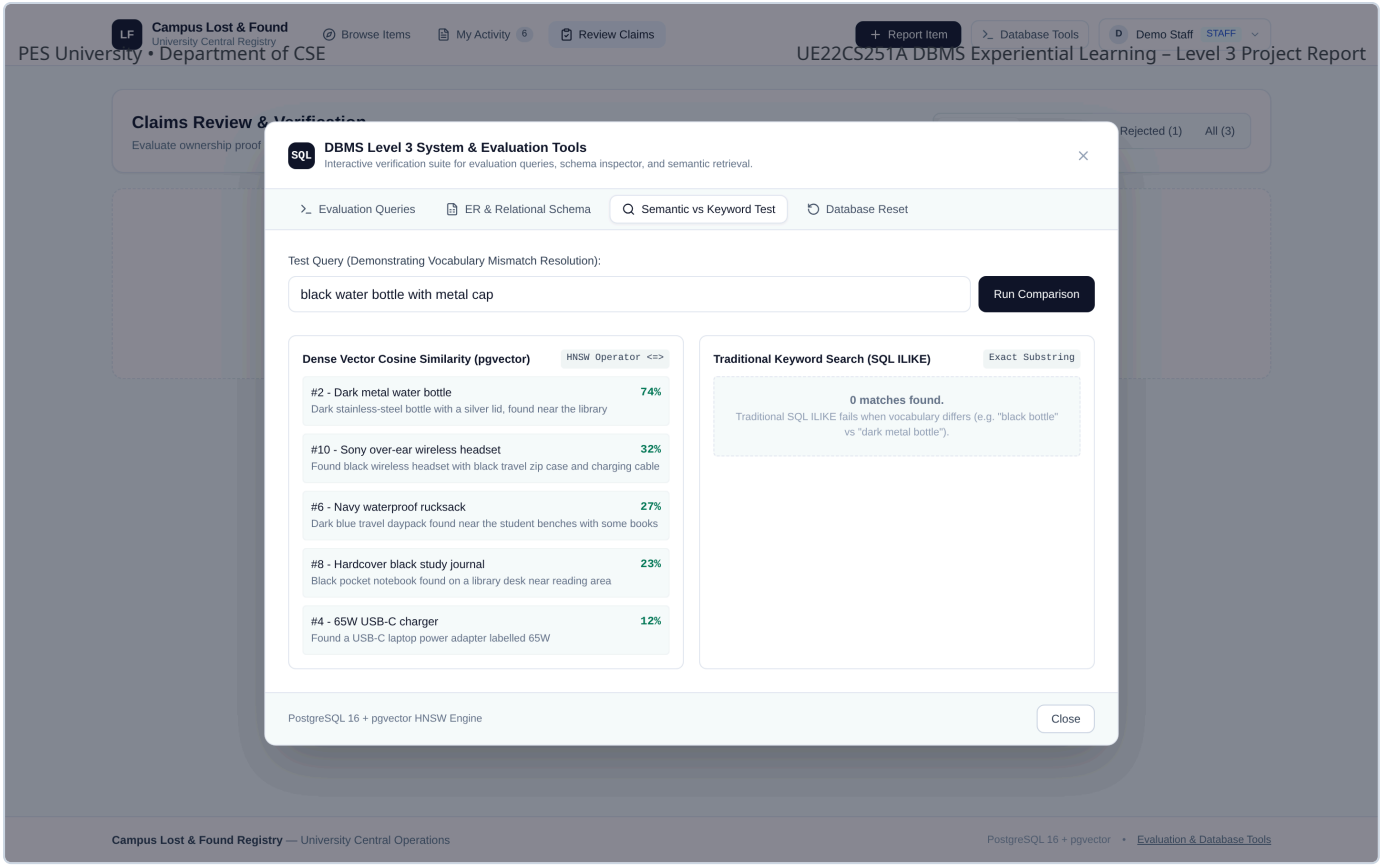


Figure 7.11: Side-by-side retrieval test demonstrating semantic vector similarity resolving vocabulary mismatch where keyword search fails.


Campus Lost & Found
University Central Registry
Browse Items
My Activity 6
Review Claims
Report Item
Database Tools
Demo Staff STAFF
Claims Review & Verification
Evaluate ownership proof
Rejected (1)
All (3)
DBMS Level 3 System & Evaluation Tools
Interactive verification suite for evaluation queries, schema inspector, and semantic retrieval.
Evaluation Queries
ER & Relational Schema
Semantic vs Keyword Test
Database Reset
Test Query (Demonstrating Vocabulary Mismatch Resolution):
black water bottle with metal cap
Run Comparison
Dense Vector Cosine Similarity (pgvector)
HNSW Operator <=>
#2 - Dark metal water bottle
74%
Dark stainless-steel bottle with a silver lid, found near the library
#10 - Sony over-ear wireless headset
32%
Found black wireless headset with black travel zip case and charging cable
#6 - Navy waterproof rucksack
27%
Dark blue travel daypack found near the student benches with some books
#8 - Hardcover black study journal
23%
Black pocket notebook found on a library desk near reading area
#4 - 65W USB-C charger
12%
Found a USB-C laptop power adapter labelled 65W
Traditional Keyword Search (SQL ILIKE)
Exact Substring
0 matches found.
Traditional SQL ILIKE fails when vocabulary differs (e.g. "black bottle" vs "dark metal bottle").
PostgreSQL 16 + pgvector HNSW Engine
Close
Campus Lost & Found Registry — University Central Operations
PostgreSQL 16 + pgvector • Evaluation & Database Tools

8. Evaluation Rubrics - Completion Status

Total Marks: 50/50

#	Evaluation Rubric	Marks	Completion Status	Detailed Implementation Evidence in Project
1	**ER Diagram**	**2**	**100% Completed**	Complete Entity-Relationship model (`public/lost_found_ER.png`) modeling entities (`users` , `category` , `location` , `item` , `item_match` , `claim`), primary keys, and cardinalities. Embedded directly in the web app under the "Database Tools" modal.
2	**Relational schema with keys, relationships, and constraints**	**3**	**100% Completed**	Fully normalized 3NF relational schema (`public/lost_found_relational_schema.png`). Primary keys, composite primary keys (`item_match`), foreign keys with `ON DELETE RESTRICT` and `ON DELETE CASCADE` , domain `CHECK` constraints (`role` , `status` , `item_type` , `similarity_score`), and `UNIQUE` constraints (`email` , `category.name` , `location.name`).
3	**SQL database implementation - DDL with constraints**	**4**	**100% Completed**	Implemented in `campus_lost_found_setup.sql` . Features `CREATE TABLE` , `ALTER TABLE item ALTER COLUMN title TYPE VARCHAR(200);` , `DROP TABLE ... CASCADE;` , composite primary keys, check constraints, default values, HNSW vector index, and partial unique index (`idx_one_approved_claim_per_item`).
4	**NoSQL / Vector database creation**	**4**	**100% Completed**	PostgreSQL 16 extended with native `pgvector` (`CREATE EXTENSION IF NOT EXISTS vector;`). High-dimensional dense vector attribute `embedding VECTOR(384)` indexed with an HNSW cosine index (`USING hnsw (embedding vector_cosine_ops)`).
5	**Technology justification**	**1**	**100% Completed**	Exhaustive justification matrix contrasting PostgreSQL + pgvector against relational MySQL, document NoSQL (MongoDB), and dedicated vector databases. Justifies eliminating synchronization drift and combining ACID claim concurrency with zero-latency vector math.
6	**DML operations**	**4**	**100% Completed**	Full DML coverage: `INSERT` (item reporting, claim filing with proof), `UPDATE` (claim status modification, item status mutation), `DELETE ... RETURNING` (item deletion with foreign key cascade), and rich multi-predicate `SELECT` queries across normalized tables.
7	**Nested / correlated / Recursive Queries, Join SQL queries**	**4**	**100% Completed**	1. **Multi-Table Join**: 4-way join across `item` , `category` , `location` , `users` . 2. **Nested Subquery**: `WHERE user_id IN (SELECT claimed_by FROM claim GROUP BY claimed_by HAVING COUNT(*) >= 1)` . 3. **Correlated Subquery**: Scalar subquery referencing outer row with `EXISTS` . 4. **Recursive Query**: `WITH RECURSIVE location_hierarchy` computing campus zone paths, depth levels, and aggregate item distribution.
8	**Advanced SQL concepts (Procedure/Function, trigger, CTE)**	**6**	**100% Completed**	1. **Stored PL/pgSQL Function**: `find_matching_items(lost_id, limit)` encapsulating vector cosine math. 2. **Triggers (3)**: `sync_item_status_after_claim` (atomic status mutation), `validate_claim_item` (enforces claims on OPEN FOUND items only), `validate_item_match_types` (validates match pair types). 3. **CTE**: Multi-stage CTE categorizing match confidence tiers (`HIGH` , `MEDIUM` , `LOW`). 4. **Views (2)**: `item_details_view` and `pending_claims_view` . 5. **Transactions**: Explicit `BEGIN ... COMMIT` with `FOR UPDATE` row-level locks.
9	**NoSQL / Vector feature, CRUD**	**7**	**100% Completed**	Full vector lifecycle: **Create** (384-d FastEmbed generation and vector storage), **Read** (pgvector cosine retrieval `<=>`), **Update** (`refreshItemMatches()` via `CROSS JOIN LATERAL find_matching_items(...)`), **Delete** (automatic cascade deletion), and sub-millisecond HNSW indexing.
10	**Next.js interface**	**3**	**100% Completed**	Responsive institutional web application built with Next.js 14 App Router, TypeScript, and Tailwind CSS. Features clean browse catalog, personal activity tracking, staff claims adjudication, item detail drawer with potential matches, and system evaluation tools.
11	**Backend / API and database integration**	**2**	**100% Completed**	Next.js API Route Handlers (`/api/items` , `/api/claims` , `/api/matches` , `/api/queries/run` , `/api/search/semantic` , `/api/meta` , `/api/demo/reset`) integrated via connection pooling (`node-postgres`). Parameterized queries protect against SQL injection. Environment variables managed in `.env.local` .
12	**Authentication implementation & RBAC (Minimum two roles)**	**4**	**100% Completed**	3 institutional roles (`STUDENT` , `STAFF` , `ADMIN`) validated by database `CHECK (role IN ('STUDENT','STAFF','ADMIN'))` and pgcrypto bcrypt hashes. Frontend session

#	Evaluation Rubric	Marks	Completion Status	Detailed Implementation Evidence in Project
				switcher modifies capabilities; backend API enforces RBAC by blocking students with `403 Forbidden` from claim adjudication.
13	**Documentation, Git / GitHub**	**2**	**100% Completed**	Clean Git repository structure, meaningful commit history, complete `README.md` with setup and live demonstration steps, and formal `PROJECT_REPORT.md` covering all 8 specified sections.
14	**Originality, individual contribution, Modification / Viva**	**4**	**100% Completed**	Solves authentic university lost-and-found vocabulary mismatch problem. Complete individual contribution by Uttam K R (PES1UG24CS697) covering both relational database engineering (schema, triggers, views, recursive CTE) and full-stack vector application architecture. Includes interactive SQL console ready for ad-hoc examiner queries and live viva modifications.

Viva Preparation & Examiner Modification Reference Guide

Potential Examiner Viva Question / Modification	System Implementation & Defense
"Show me your Recursive Query and explain how it terminates."	Open **Database Tools** -> **Evaluation Queries** -> Select **Query 4: Recursive Location Hierarchy**. The anchor member selects root zones (`parent_location_id IS NULL`), while the recursive member joins child zones on `parent_location_id = lh.location_id` . It terminates when no more child zones exist in the table.
"Prove that your database trigger works inside an ACID transaction."	In the application, navigate to **Review Claims** (as Staff or Admin) -> Click **Approve** on Claim #1. A transaction banner confirms that `sync_item_status_after_claim` fired inside `BEGIN...COMMIT` , mutating the item's status from `OPEN` to `CLAIMED` .
"What happens if a student attempts to file a claim on a LOST item?"	Run Query 13 in the console. The database trigger `validate_claim_item` raises an exception: *"Claims can only be submitted for FOUND items"*, aborting the transaction immediately.
"How does your vector search outperform keyword matching?"	Open **Database Tools** -> **Semantic vs Keyword Test**. Search for *"black water bottle with metal cap"*. The vector engine returns *"Dark metal water bottle"* with ~0.74 cosine similarity, while keyword `ILIKE` returns zero matches due to vocabulary mismatch.
"Can two claims for the same item both be approved?"	No. Partial unique index `CREATE UNIQUE INDEX idx_one_approved_claim_per_item ON claim(item_id) WHERE claim_status = 'APPROVED';` enforces at the database engine level that duplicate approved claims are rejected.